Capstone Exercises

Decision Modeling for Health Economic Evaluation Workshop

Cebu, Philippines 2026

Capstone Exercise 1: Specify the Decision Problem

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Capstone Exercise 1
Specify the Decision Problem

- Objectives
  - Present your draft Background, Objective/Decision Problem, and Strategy Options slides to your small group
  - Provide feedback on clarity and completeness of descriptions
    1. How can the health issue/condition be more clearly described?
    2. Is the decision problem clearly defined?
    3. How well do the strategy options match the background and decision problem?
- Deliverables
  - Refined slide descriptions of background, decision problem, and strategies based on feedback
    - Slide 2 – Introduction/Background
    - Slide 3 – Objective/Decision problem
    - Slide 4 – Strategy options

Capstone Exercise 2: Build a Decision Tree

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Capstone Exercise 2, morning exercise

Build a Decision Tree

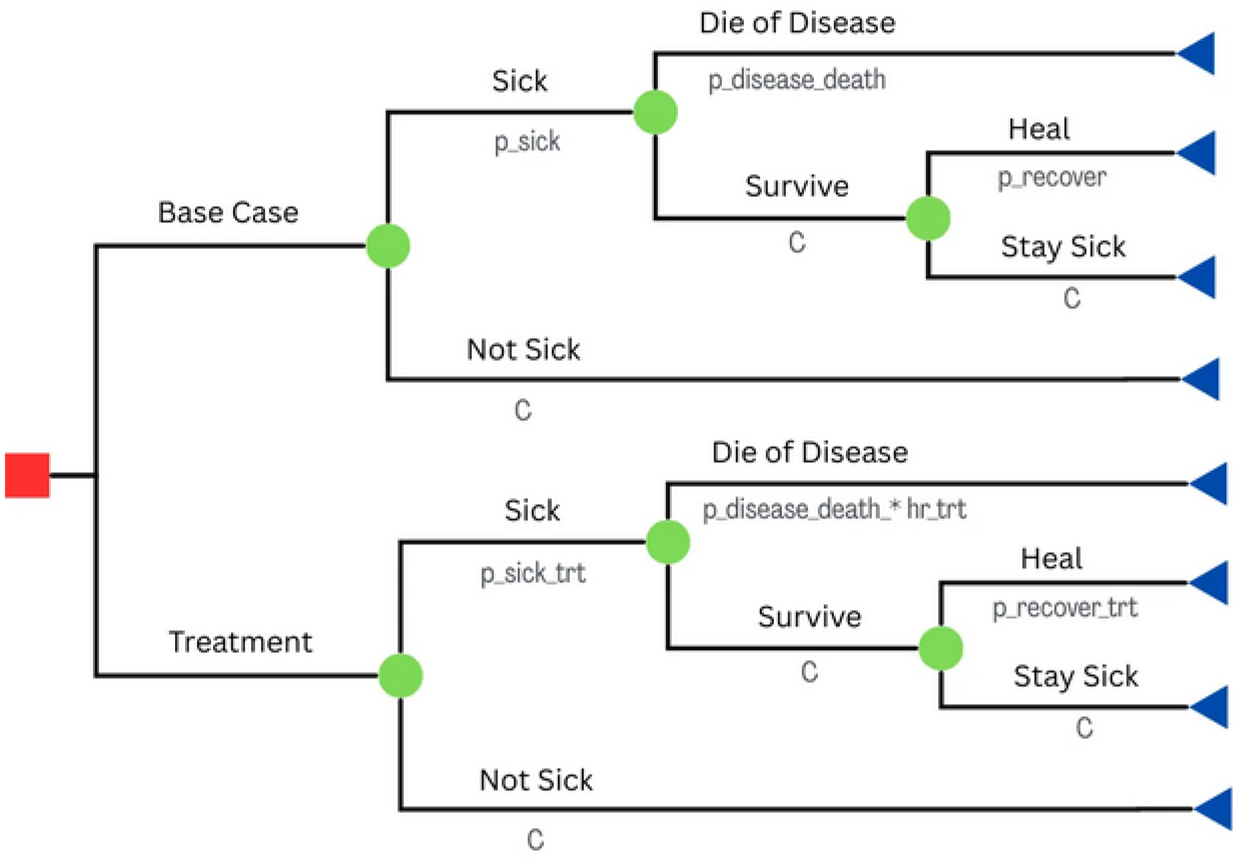
- Objectives
  - Construct a decision tree in Amua comparing the status quo to your specific strategy(ies)

- Deliverables
  - Begin a slide for final presentation that graphically displays your decision options using a decision tree
    - Slide 5 – Decision tree graphic

Capstone Exercise 2

Build a Decision Tree

Example



Capstone Exercise 2: Parameterize Decision Tree

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Capstone Exercise 2, afternoon exercise

Add Parameters to a Decision Tree

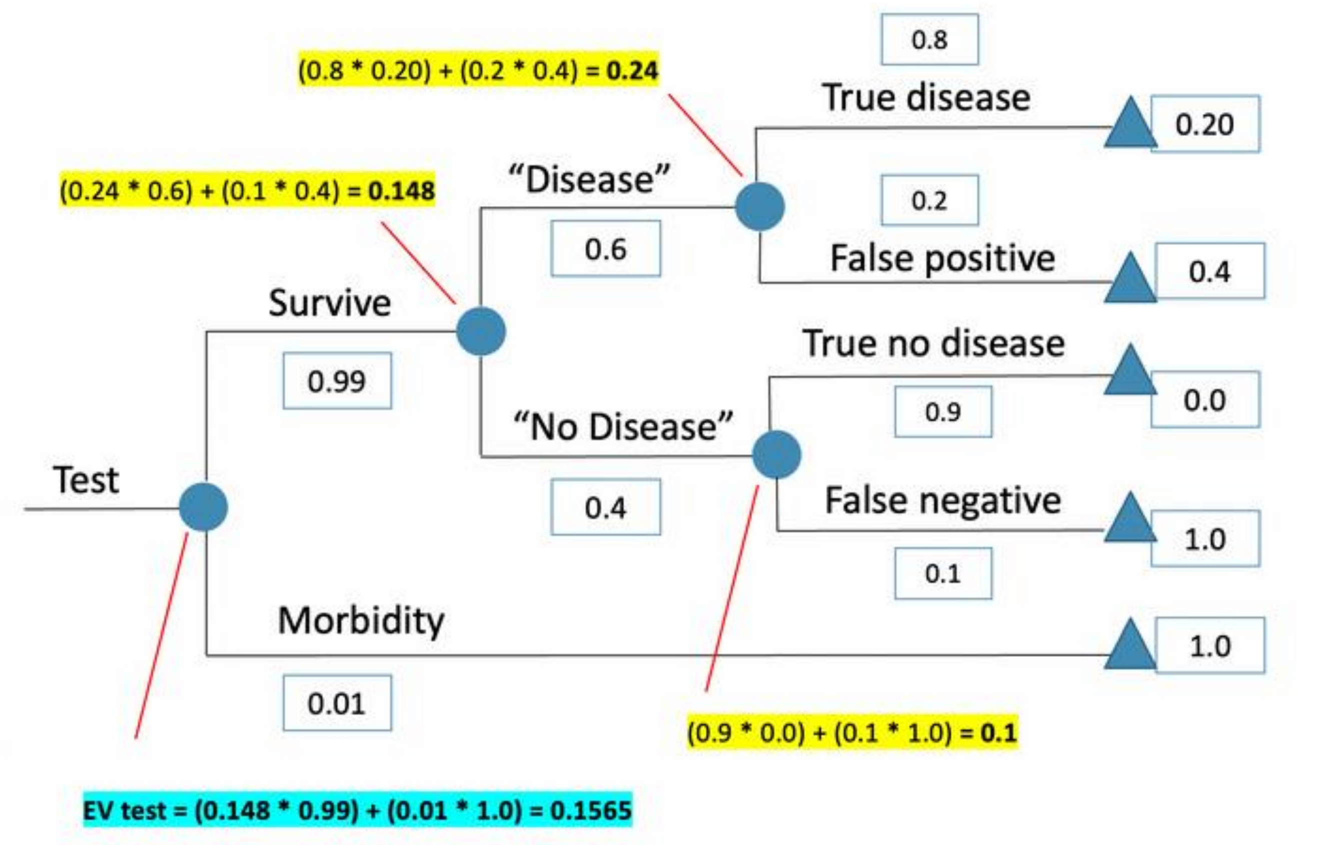
- Objectives
  - Add parameters to your decision tree and estimate outcomes

- Deliverables
  - Revise slide for final presentation that graphically displays your decision options using a decision tree with outcomes
    - Slide 5 – Decision tree graphic with expected values and outcomes

Capstone Exercise 2

Build a Decision Tree

Example



Capstone Exercise 3: Design the Markov Model

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Capstone Exercise 3

Design a Markov Model

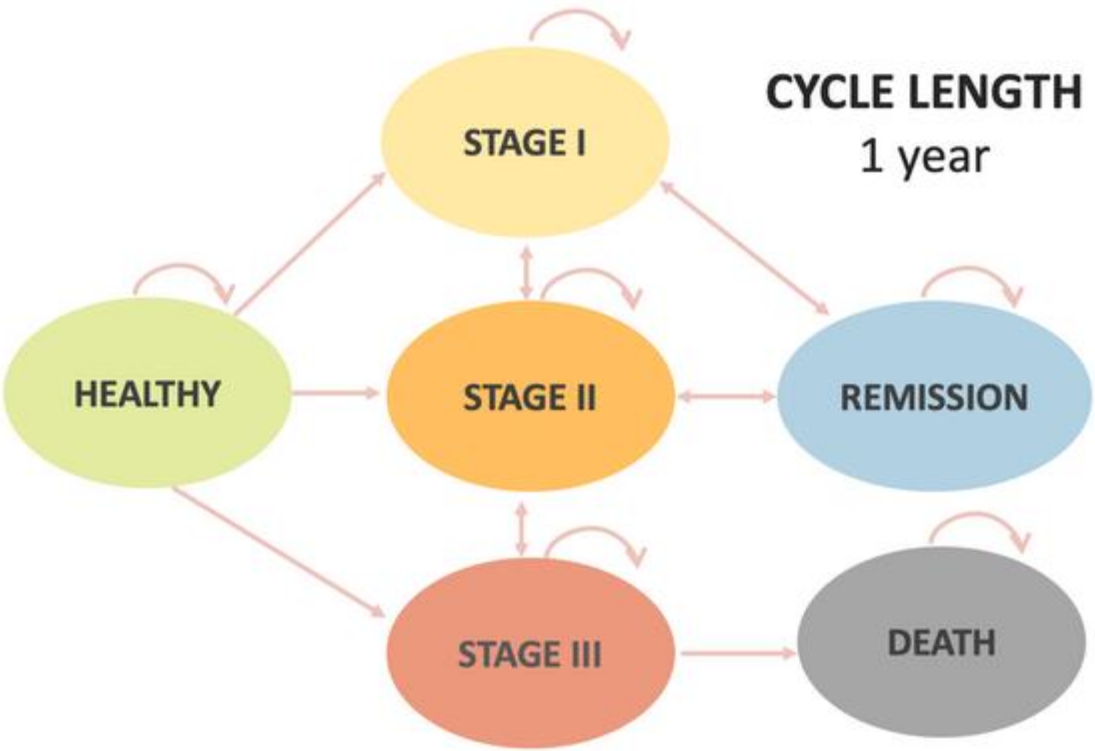
- Objective
  - Use your decision problem and strategies from slides 3, 4, and 5
    - Determine the health states of the decision problem
    - Determine the transitions
    - Determine parameters
      - Population
      - Cycle length
      - Time horizon
      - Probabilities (fill in later)

- Deliverable
  - Create slide for final presentation that displays your decision problem as a Markov schematic
    - Slide 6 – bubble diagram graphic, transition matrix framework, parameter table

Capstone Exercise 3

Design a Markov Model

Examples



Parameter Table

param	Base_case	description	notes	distribution	source
n_age_init		Age at baseline			
n_age_max		Maximum age of followup			
c_S		Annual cost of sick(S)			

Transition Matrix

	Healthy	Stage I	Stage II	Stage III	Remission	Death
Healthy						
Stage I	0					
Stage II	0	0				
Stage III	0	0	0			
Remission	0	0	0	0		
Death	0	0	0	0	0	1

Capstone Exercise 4: Run the Markov Model

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Capstone Exercise 4

Run the Markov Model

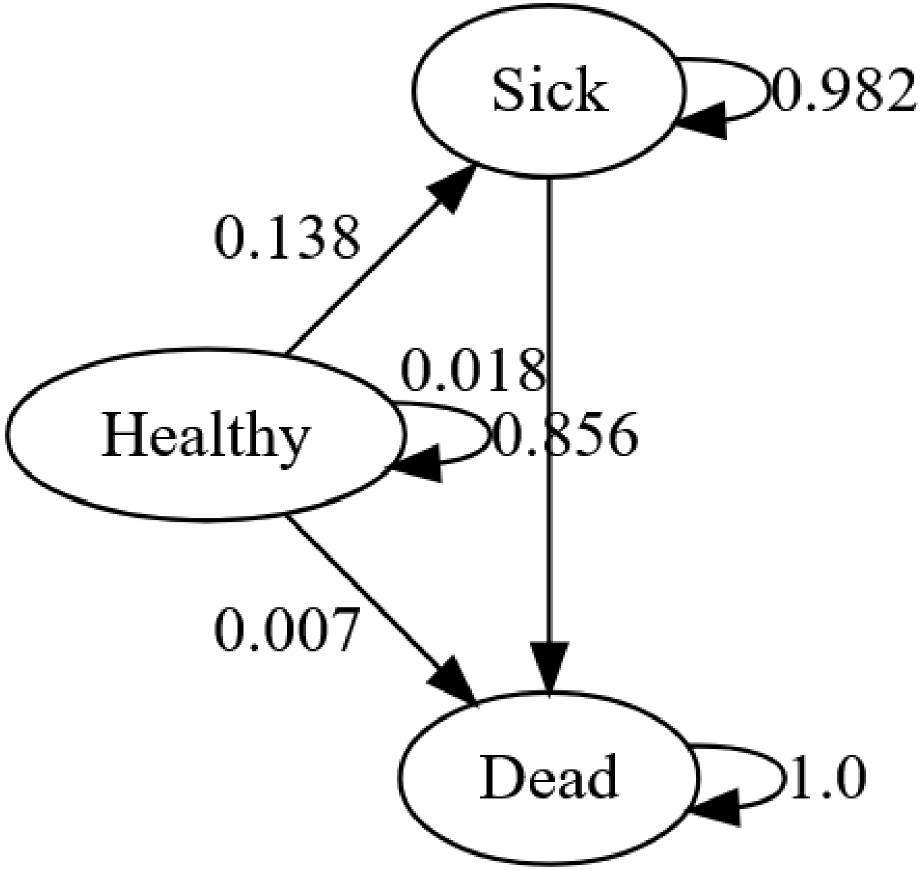
- Objective
  - Run Markov model in Amua
  - Complete a Markov trace table for *one* strategy

- Deliverable
  - Create slides for final presentation that fills in and documents your Markov model inputs and calculation
    - Slide 7 – Markov bubble schematic, “reward” calculations, and outcome

Capstone Exercise 4

Run the Markov Model

Examples



Health State Occupancy Over Ten Cycles

cycle	H	S	D
0	1.00000	0.00000	0.000000
1	0.85600	0.13800	0.007000
2	0.73274	0.25364	0.015476
3	0.62722	0.35020	0.025171
4	0.53690	0.43045	0.035865
5	0.45959	0.49679	0.047371
6	0.39341	0.55127	0.059531
7	0.33676	0.59564	0.072207
8	0.28826	0.63139	0.085286
9	0.24675	0.65981	0.098669
10	0.21122	0.68198	0.112273